

Additional Topics

4.A Basepoints and Homotopy

In the first part of this section we will use the action of π_1 on π_n to describe the difference between $\pi_n(X, x_0)$ and the set of homotopy classes of maps $S^n \rightarrow X$ without conditions on basepoints. More generally, we will compare the set $\langle Z, X \rangle$ of basepoint-preserving homotopy classes of maps $(Z, z_0) \rightarrow (X, x_0)$ with the set $[Z, X]$ of unrestricted homotopy classes of maps $Z \rightarrow X$, for Z any CW complex with basepoint z_0 a 0-cell. Then the section concludes with an extended example exhibiting some rather subtle nonfinite generation phenomena in homotopy and homology groups.

We begin by constructing an action of $\pi_1(X, x_0)$ on $\langle Z, X \rangle$ when Z is a CW complex with basepoint 0-cell z_0 . Given a loop γ in X based at x_0 and a map $f_0: (Z, z_0) \rightarrow (X, x_0)$, then by the homotopy extension property there is a homotopy $f_s: Z \rightarrow X$ of f_0 such that $f_s(z_0)$ is the loop γ . We might try to define an action of $\pi_1(X, x_0)$ on $\langle Z, X \rangle$ by $[\gamma][f_0] = [f_1]$, but this definition encounters a small problem when we compose loops. For if η is another loop at x_0 , then by applying the homotopy extension property a second time we get a homotopy of f_1 restricting to η on x_0 , and the two homotopies together give the relation $([\gamma][\eta])[f_0] = [\eta]([\gamma][f_0])$, in view of our convention that the product $\gamma\eta$ means first γ , then η . This is not quite the relation we want, but the problem is easily corrected by letting the action be an action on the right rather than on the left. Thus we set $[f_0][\gamma] = [f_1]$, and then $[f_0]([\gamma][\eta]) = ([f_0][\gamma])[\eta]$.

Let us check that this right action is well-defined. Suppose we start with maps $f_0, g_0: (Z, z_0) \rightarrow (X, x_0)$ representing the same class in $\langle Z, X \rangle$, together with homotopies f_s and g_s of f_0 and g_0 such that $f_s(z_0)$ and $g_s(z_0)$ are homotopic loops. These various homotopies define a map $H: Z \times I \times \partial I \cup Z \times \{0\} \times I \cup \{z_0\} \times I \times I \rightarrow X$ which is f_s on $Z \times I \times \{0\}$, g_s on $Z \times I \times \{1\}$, the basepoint-preserving homotopy from f_0 to g_0 on $Z \times \{0\} \times I$, and the homotopy from $f_s(z_0)$ to $g_s(z_0)$ on $\{z_0\} \times I \times I$. We would like to extend H over $Z \times I \times I$. The pair $(I \times I, I \times \partial I \cup \{0\} \times I)$ is homeomorphic to $(I \times I, I \times \{0\})$, and via this homeomorphism we can view H as a map $Z \times I \times \{0\} \cup \{z_0\} \times I \times I \rightarrow X$, that is, a map $Z \times I \rightarrow X$ with a homotopy on the subcomplex $\{z_0\} \times I$. This means the homotopy extension property can be applied to produce an extension of the original H to $Z \times I \times I$. Restricting this extended H to $Z \times \{1\} \times I$ gives a basepoint-preserving homotopy $f_1 \simeq g_1$, which shows that $[f_0][\gamma]$ is well-defined.

Note that in this argument we did not have to assume the homotopies f_s and g_s were constructed by applying the homotopy extension property. Thus we have proved

the following result:

Proposition 4A.1. *There is a right action of $\pi_1(X, x_0)$ on $\langle Z, X \rangle$ defined by setting $[f_0][\gamma] = [f_1]$ whenever there exists a homotopy $f_s: Z \rightarrow X$ from f_0 to f_1 such that $f_s(z_0)$ is the loop γ , or any loop homotopic to γ . $\square$*

It is easy to convert this right action into a left action, by defining $[\gamma][f_0] = [f_0][\gamma]^{-1}$. This just amounts to choosing the homotopy f_s so that $f_s(z_0)$ is the inverse path of γ .

When $Z = S^n$ this action reduces to the usual action of $\pi_1(X, x_0)$ on $\pi_n(X, x_0)$ since in the original definition of γf in terms of maps $(I^n, \partial I^n) \rightarrow (X, x_0)$, a homotopy from γf to f is obtained by restricting γf to smaller and smaller concentric cubes, and on the 'basepoint' ∂I^n this homotopy traces out the loop γ .

Proposition 4A.2. *If (Z, z_0) is a CW pair and X is a path-connected space, then the natural map $\langle Z, X \rangle \rightarrow [Z, X]$ induces a bijection of the orbit set $\langle Z, X \rangle / \pi_1(X, x_0)$ onto $[Z, X]$.*

In particular, this implies that $[Z, X] = \langle Z, X \rangle$ if X is simply-connected.

Proof: Since X is path-connected, every $f: Z \rightarrow X$ can be homotoped to take z_0 to the basepoint x_0 , via homotopy extension, so the map $\langle Z, X \rangle \rightarrow [Z, X]$ is onto. If f_0 and f_1 are basepoint-preserving maps that are homotopic via the homotopy $f_s: Z \rightarrow X$, then by definition $[f_1] = [f_0][\gamma]$ for the loop $\gamma(s) = f_s(z_0)$, so $[f_0]$ and $[f_1]$ are in the same orbit under the action of $\pi_1(X, x_0)$. Conversely, two basepoint-preserving maps in the same orbit are obviously homotopic. $\square$

Example 4A.3. If X is an H-space with identity element x_0 , then the action of $\pi_1(X, x_0)$ on $\langle Z, X \rangle$ is trivial since for a map $f: (Z, z_0) \rightarrow (X, x_0)$ and a loop γ in X based at x_0 , the multiplication in X defines a homotopy $f_s(z) = f(z)\gamma(s)$. This starts and ends with a map homotopic to f , and the loop $f_s(z_0)$ is homotopic to γ , both these homotopies being basepoint-preserving by the definition of an H-space.

The set of orbits of the π_1 action on π_n does not generally inherit a group structure from π_n . For example, when $n = 1$ the orbits are just the conjugacy classes in π_1 , and these form a group only when π_1 is abelian. Basepoints are thus a necessary technical device for producing the group structure in homotopy groups, though as we have shown, they can be ignored in simply-connected spaces.

For a set of maps $S^n \rightarrow X$ to generate $\pi_n(X)$ as a module over $\mathbb{Z}[\pi_n(X)]$ means that all elements of $\pi_n(X)$ can be represented by sums of these maps along arbitrary paths in X , where we allow reversing orientations to get negatives and repetitions to get arbitrary integer multiples. Examples of finite CW complexes X for which $\pi_n(X)$ is not finitely generated as a module over $\mathbb{Z}[\pi_n(X)]$ were given in Exercise 38 in §4.2, provided $n \geq 3$. Finding such an example for $n = 2$ seems to be more difficult. The

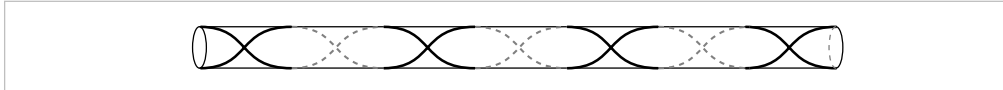
rest of this section will be devoted to a somewhat complicated construction which does this, and is interesting for other reasons as well.

An Example of Nonfinite Generation

We will construct a finite CW complex having π_n not finitely generated as a $\mathbb{Z}[\pi_1]$ -module, for a given integer $n \geq 2$. The complex will be a subcomplex of a $K(\pi, 1)$ having interesting homological properties: It is an $(n + 1)$ -dimensional CW complex with H_{n+1} nonfinitely generated, but its n -skeleton is finite so H_i is finitely generated for $i \leq n$ and π is finitely presented if $n > 1$. The first such example was found in [Stallings 1963] for $n = 2$. Our construction will be essentially the n -dimensional generalization of this, but described in a more geometric way as in [Bestvina & Brady 1997], which provides a general technique for constructing many examples of this sort.

To begin, let X be the product of n copies of $S^1 \vee S^1$. Since $S^1 \vee S^1$ is the 1-skeleton of the torus $T^2 = S^1 \times S^1$ in its usual CW structure, X can be regarded as a subcomplex of the $2n$ -dimensional torus T^{2n} , the product of $2n$ circles. Define $f: T^{2n} \rightarrow S^1$ by $f(\theta_1, \dots, \theta_{2n}) = \theta_1 + \dots + \theta_{2n}$ where the coordinates $\theta_i \in S^1$ are viewed as angles measured in radians. The space $Z = X \cap f^{-1}(0)$ will provide the example we are looking for. As we shall see, Z is a finite CW complex of dimension $n - 1$, with $\pi_{n-1}(Z)$ nonfinitely generated as a module over $\pi_1(Z)$ if $n \geq 3$. We will also see that $\pi_i(Z) = 0$ for $1 < i < n - 1$.

The induced homomorphism $f_*: \pi_1(T^{2n}) \rightarrow \pi_1(S^1) = \mathbb{Z}$ sends each generator coming from an S^1 factor to 1. Let $\tilde{T}^{2n} \rightarrow T^{2n}$ be the covering space corresponding to the kernel of f_* . This is a normal covering space since it corresponds to a normal subgroup, and the deck transformation group is $\mathbb{Z}$. The subcomplex of $\tilde{T}^{2n}$ projecting to X is a normal covering space $\tilde{X} \rightarrow X$ with the same group of deck transformations. Since $\pi_1(X)$ is the product of n free groups on two generators, $\tilde{X}$ is the covering space of X corresponding to the kernel of the homomorphism $\pi_1(X) \rightarrow \mathbb{Z}$ sending each of the two generators of each free factor to 1. Since X is a $K(\pi, 1)$, so is $\tilde{X}$. For example, when $n = 1$, $\tilde{X}$ is the union of two helices on the infinite cylinder $\tilde{T}^2$:



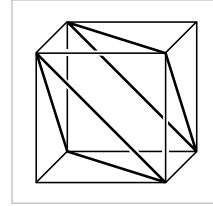
The map f lifts to a map $\tilde{f}: \tilde{T}^{2n} \rightarrow \mathbb{R}$, and Z lifts homeomorphically to a subspace $Z \subset \tilde{X}$, namely $\tilde{f}^{-1}(0) \cap \tilde{X}$. We will show:

- (*) $\tilde{X}$ is homotopy equivalent to a space Y obtained from Z by attaching an infinite sequence of n -cells.

Assuming this is true, it follows that $H_n(Y)$ is not finitely generated since in the exact sequence $H_n(Z) \rightarrow H_n(Y) \rightarrow H_n(Y, Z) \rightarrow H_{n-1}(Z)$ the first term is zero and the last term is finitely generated, Z being a finite CW complex of dimension $n - 1$,

while the third term is an infinite sum of $\mathbb{Z}$'s, one for each n -cell of Y . If $\pi_{n-1}(Z)$ were finitely generated as a $\pi_1(Z)$ -module, then by attaching finitely many n -cells to Z we could make it $(n-1)$ -connected since it is already $(n-2)$ -connected as the $(n-1)$ -skeleton of the $K(\pi, 1)$ Y . Then by attaching cells of dimension greater than n we could build a $K(\pi, 1)$ with finite n -skeleton. But this contradicts the fact that $H_n(Y)$ is not finitely generated.

To begin the verification of $(*)$, consider the torus T^m . The standard cell structure on T^m lifts to a cubical cell structure on the universal cover $\mathbb{R}^m$, with vertices the integer lattice points $\mathbb{Z}^m$. The function f lifts to a linear projection $L: \mathbb{R}^m \rightarrow \mathbb{R}$, $L(x_1, \dots, x_m) = x_1 + \dots + x_m$. The planes in $L^{-1}(\mathbb{Z})$ cut the cubes of $\mathbb{R}^m$ into convex polyhedra which we call *slabs*. There are m slabs in each m -dimensional cube. The boundary of a slab in $L^{-1}[i, i+1]$ consists of lateral faces that are slabs for lower-dimensional cubes, together with a lower face in $L^{-1}(i)$ and an upper face in $L^{-1}(i+1)$. In each cube there are two exceptional slabs whose lower or upper face degenerates to a point. These are the slabs containing the vertices of the cube where L has its maximum and minimum values. A slab deformation retracts onto the union of its lower and lateral faces, provided that the slab has an upper face that is not just a point. Slabs of the latter type are m -simplices, and we will refer to them as *cones* in what follows. These are the slabs containing the vertex of a cube on which L takes its maximal value. The lateral faces of a cone are also cones, of lower dimension.



The slabs, together with all their lower-dimensional faces, give a CW structure on $\mathbb{R}^m$ with the planes of $L^{-1}(\mathbb{Z})$ as subcomplexes. These structures are preserved by the deck transformations of the cover $\mathbb{R}^m \rightarrow T^m$ so there is an induced CW structure in the quotient T^m , with $f^{-1}(0)$ as a subcomplex.

If X is any subcomplex of T^m in its original cubical cell structure, then the slab CW structure on T^m restricts to a CW structure on X . In particular, we obtain a CW structure on $Z = X \cap f^{-1}(0)$. Likewise we get a lifted CW structure on the cover $\tilde{X} \subset \tilde{T}^m$. Let $\tilde{X}[i, j] = \tilde{X} \cap \tilde{f}^{-1}[i, j]$. The deformation retractions of noncone slabs onto their lateral and lower faces give rise to a deformation retraction of $\tilde{X}[i, i+1]$ onto $\tilde{X}[i] \cup C_i$ where C_i consists of all the cones in $\tilde{X}[i, i+1]$. These cones are attached along their lower faces, and they all have the same vertex in $\tilde{X}[i+1]$, so C_i is itself a cone in the usual sense, attached to $\tilde{X}[i]$ along its base.

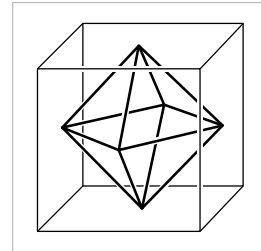
For the particular X we are interested in, we claim that each C_i is an n -disk attached along its boundary sphere. When $n = 1$ this is evident from the earlier picture of $\tilde{X}$ as the union of two helices on a cylinder. For larger n we argue by induction. Passing from n to $n+1$ replaces X by two copies of $X \times S^1$ intersecting in X , one copy for each of the additional S^1 factors of T^{2n+2} . Replacing X by $X \times S^1$ changes C_i to its join with a point in the base of the new C_i . Doing this twice produces

the suspension of C_i attached along the suspension of the base.

The same argument shows that $\tilde{X}[-i-1, -i]$ deformation retracts onto $\tilde{X}[-i]$ with an n -cell attached. We build the space Y and a homotopy equivalence $g: Y \rightarrow \tilde{X}$ by an inductive procedure, starting with $Y_0 = Z$. Assuming that Y_i and a homotopy equivalence $g_i: Y_i \rightarrow \tilde{X}[-i, i]$ have already been defined, we form Y_{i+1} by attaching two n -cells by the maps obtained from the attaching maps of the two n -cells in $\tilde{X}[-i-1, i+1] - \tilde{X}[-i, i]$ by composing with a homotopy inverse to g_i . This allows g_i to be extended to a homotopy equivalence $g_{i+1}: Y_{i+1} \rightarrow \tilde{X}[-i-1, i+1]$. Taking the union over i gives $g: Y \rightarrow \tilde{X}$. One can check this is a homotopy equivalence by seeing that it induces isomorphisms on all homotopy groups, using the standard compactness argument. This finishes the verification of (*).

It is interesting to see what the complex Z looks like in the case $n = 3$, when Z is 2-dimensional and has π_2 nonfinitely generated over $\mathbb{Z}[\pi_1(Z)]$. In this case X is the product of three $S^1 \vee S^1$'s, so X is the union of the eight 3-tori obtained by choosing one of the two S^1 summands in each $S^1 \vee S^1$ factor. We denote these 3-tori $S^1_{\pm} \times S^1_{\pm} \times S^1_{\pm}$. Viewing each of these 3-tori as the cube in the previous figure with opposite faces identified, we see that Z is the union of the eight 2-tori formed by the two sloping triangles in each cube. Two of these 2-tori intersect along a circle when the corresponding 3-tori of X intersect along a 2-torus. This happens when the triples of $\pm$'s for the two 3-tori differ in exactly one entry. The pattern of intersection of the eight 2-tori of Z can thus be described combinatorially via the 1-skeleton of the cube, with vertices $(\pm 1, \pm 1, \pm 1)$. There is a torus of Z for each vertex of the cube, and two tori intersect along a circle when the corresponding vertices of the cube are the endpoints of an edge of the cube. All eight tori contain the single 0-cell of Z .

To obtain a model of Z itself, consider a regular octahedron inscribed in the cube with vertices $(\pm 1, \pm 1, \pm 1)$. If we identify each pair of opposite edges of the octahedron, each pair of opposite triangular faces becomes a torus. However, there are only four pairs of opposite faces, so we get only four tori this way, not eight. To correct this problem, regard each triangular face of the octahedron as two copies of the same triangle, distinguished from each other by a choice of normal direction, an arrow attached to the triangle pointing either inside the octahedron or outside it, that is, either toward the nearest vertex of the surrounding cube or toward the opposite vertex of the cube. Then each pair of opposite triangles of the octahedron having normal vectors pointing toward the same vertex of the cube determines a torus, when opposite edges are identified as before. Each edge of the original octahedron is also replaced by two edges oriented either toward the interior or exterior of the octahedron. The vertices of the octahedron may be left unduplicated since they will all be identified to a single point anyway. With this scheme, the two tori corresponding to the vertices at the ends



of an edge of the cube then intersect along a circle, as they should, and other pairs of tori intersect only at the 0-cell of Z .

This model of Z has the advantage of displaying the symmetry group of the cube, a group of order 48, as a symmetry group of Z , corresponding to the symmetries of X permuting the three $S^1 \vee S^1$ factors and the two S^1 's of each $S^1 \vee S^1$. Undoubtedly Z would be very pretty to look at if we lived in a space with enough dimensions to see all of it at one glance.

It might be interesting to see an explicit set of maps $S^2 \rightarrow Z$ generating $\pi_2(Z)$ as a $\mathbb{Z}[\pi_1]$ -module. One might also ask whether there are simpler examples of these nonfinite generation phenomena.

Exercises

1. Show directly that if X is a topological group with identity element x_0 , then any two maps $f, g: (Z, z_0) \rightarrow (X, x_0)$ which are homotopic are homotopic through basepoint-preserving maps.
2. Show that under the map $\langle X, Y \rangle \rightarrow \text{Hom}(\pi_n(X, x_0), \pi_n(Y, y_0))$, $[f] \mapsto f_*$, the action of $\pi_1(Y, y_0)$ on $\langle X, Y \rangle$ corresponds to composing with the action on $\pi_n(Y, y_0)$, that is, $(\gamma f)_* = \beta_\gamma f_*$. Deduce a bijection of $[X, K(\pi, 1)]$ with the set of orbits of $\text{Hom}(\pi_1(X), \pi)$ under composition with inner automorphisms of π . In particular, if π is abelian then $[X, K(\pi, 1)] = \langle X, K(\pi, 1) \rangle = \text{Hom}(\pi_1(X), \pi)$.
3. For a space X let $\text{Aut}(X)$ denote the group of homotopy classes of homotopy equivalences $X \rightarrow X$. Show that for a CW complex $K(\pi, 1)$, $\text{Aut}(K(\pi, 1))$ is isomorphic to the group of outer automorphisms of π , that is, automorphisms modulo inner automorphisms.
4. With the notation of the preceding problem, show that $\text{Aut}(\bigvee_n S^k) \approx \text{GL}_n(\mathbb{Z})$ for $k > 1$, where $\bigvee_n S^k$ denotes the wedge sum of n copies of S^k and $\text{GL}_n(\mathbb{Z})$ is the group of $n \times n$ matrices with entries in $\mathbb{Z}$ having an inverse matrix of the same form. [$\text{GL}_n(\mathbb{Z})$ is the automorphism group of $\mathbb{Z}^n \approx \pi_k(\bigvee_n S^k) \approx H_k(\bigvee_n S^k)$.]
5. This problem involves the spaces constructed in the latter part of this section.
 - (a) Compute the homology groups of the complex Z in the case $n = 3$, when Z is 2-dimensional.
 - (b) Letting $\tilde{X}_n$ denote the n -dimensional complex $\tilde{X}$, show that $\tilde{X}_n$ can be obtained inductively from $\tilde{X}_{n-1}$ as the union of two copies of the mapping torus of the generating deck transformation $\tilde{X}_{n-1} \rightarrow \tilde{X}_{n-1}$, with copies of $\tilde{X}_{n-1}$ in these two mapping tori identified. Thus there is a fiber bundle $\tilde{X}_n \rightarrow S^1 \vee S^1$ with fiber $\tilde{X}_{n-1}$.
 - (c) Use part (b) to find a presentation for $\pi_1(\tilde{X}_n)$, and show this presentation reduces to a finite presentation if $n > 2$ and a presentation with a finite number of generators if $n = 2$. In the latter case, deduce that $\pi_1(\tilde{X}_2)$ has no finite presentation from the fact that $H_2(\tilde{X}_2)$ is not finitely generated.